



**Verified Carbon
Standard**

INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN BANGLADESH

Document Prepared by
C-Quest Capital BD Stoves 1 Pte Ltd

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1 PROJECT DETAILS

1.1 Summary Description of the Project

A summary description of the technologies/ measures to be implemented by the project.

The project involves distribution of fuel-efficient improved cookstoves (ICS) in Bangladesh. The ICS disseminated through this project will replace the baseline cookstoves.

Through this project, approximately 1 million household will be undertaken to distribution and installation of ICS in Bangladesh. It is intended that under this project Baldwin adjustable portable stove, Dishari Green Stove, Baldwin double pot, and Dinabandhu stove will be distributed. Other ICS models may be added based on the requirement of end users. The ICS are designed to operate at higher efficiencies than the baseline stoves and result in reduced wood consumption as compared to baseline scenario. The overall effect is reduced emission of carbon dioxide and smoke, improving air quality within the household as well as the quality of life of women folk. That practice will halt the rapidly progressing deforestation in Bangladesh. At present the total stove distribution under the project activity is set at 1,600,000¹ of which 600,000 will be Baldwin stove (8 inch), 400,000 will be Baldwin stove (10 inch), 200,000 will be Dishari Green Stove, 200,000 will be Baldwin Double Pot and 200,000 will be Dinabandhu stove for end user households. Under the project activity, each household will receive upto two stoves either a single fixed Dinabandhu/Baldwin Double Pot Cookstove or two portable cookstoves (combination of Baldwin 8 inch, Baldwin 10 inch, and Dishari Green cookstove) as part of the distribution of the stoves. However, the number of stoves that will be given to end user will depend on various factors such as household size, it's cooking requirement etc² and the project proponent and/or the implementing partner shall have the discretion to decide on the combination as well as number of stoves each beneficiary household will receive. This is to ensure that the end user is not forced to go back to using baseline stoves on account of one project stove not being able to cater to complete requirements of the household. Also, since each of the stoves will have a specific unique identification consisting of alpha numeric or any other identification system, even if a household receives a combination of stoves, each of them can be traced easily.

The location of the project.

The project will take place in Bangladesh. The details of the project location are provided in Section 1.12.

An explanation of how the project is expected to generate GHG emission reductions or removals.

Each ICS is the project activity instance that will substitute the current practice of using the traditional mud stove or three stone fire or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. Fuelwood is burnt more

¹ This is a tentative number; final number will depend on actual stove distribution.

² Shall be at the discretion of the project proponent and/or implementor.

efficiently with the ICS because its thermal efficiency is higher, resulting in a reduction in fuel consumption and greenhouse gas emissions.

A brief description of the scenario existing prior to the implementation of the project.

The baseline scenario is the continued use of non-renewable wood fuel on traditional mud stove or three stone fired or a conventional cook stove without a grate or a chimney by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

An estimate of annual average and total GHG emission reduction and removals.

The annual average GHG emission reductions from the project is estimated to be 10,882,863 tCO₂e and the total reduction over the entire crediting period is expected to be 108,828,639 tCO₂e. And the estimated annual GHG emission reduction for each project activity instance is ~6.8 tCO₂e.

Audit Type	Period	Program	VVB Name	Number of years
Validation (Ongoing)	20-April-2023 (Date of Listing of VCS-PD)	VCS	Carbon Check (India) Pvt Ltd	-

1.2 Sectoral Scope and Project Type

The project is categorised under type/category as below:

- a) **Sectoral scope:** 03 - Energy demand
- b) **Type:** II – Energy efficiency improvement projects

The project is a grouped project.

1.3 Project Eligibility

The project abides by the principles as mentioned in Section 2.1.1 of the VCS Standard, version 4.4. The detailed justification of the eligibility in regard to the principles have been provided in the table below:

Eligibility Conditions	Justification
The six Kyoto Protocol greenhouse gases	The project aims at reducing the emission of Carbon dioxide (CO ₂), Methane (CH ₄) and Nitrous Oxide (N ₂ O), which are included in the Kyoto protocol's list of greenhouse gases

Ozone-depleting substances	Not applicable as the project activity involves the distribution of improved cookstoves
Project activities supported by a methodology approved under the VCS Program through the methodology approval process	Emissions saved by the project shall be calculated using VCS approved methodology- VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1
Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Not applicable as the project shall be using VCS approved methodology, VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1, to determine the emission reduction through the project activity
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	Not applicable as the project activity involves the distribution of improved cookstoves and hence the project does not come under the purview of jurisdictional REDD+ program

The project activity has not been undertaken primarily for the purpose of the subsequent reduction, removal, or destruction of GHG emissions and the activity of the project (i.e., distribution of efficient cookstoves) is also eligible for VCS Program, as per Table 1 in Section 2.1.3 of the VCS Standard v.4.4.

1.4 Project Design

- ☐ The project includes a single location or installation only
- ☐ The project includes multiple locations or project activity instances, but is not being developed as a grouped project
- ☒ The project is a grouped project

Eligibility Criteria

For the inclusion of new project activity instances i.e., ICS, the project proponent shall ensure that it meets the eligibility criteria below.

Sl. No	Criterion	Compliance Requirement	Method of Evaluation
1.	Methodology: Meet the applicability conditions set out in the methodology applied to the project	The project activity instance shall use VCS approved methodology - VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1 and shall comply with its requirements.	Details of how the new project activity instances meet the applicability conditions can be confirmed in Section 3.2 of this VCS-PD.
2.	Technology: Use the technologies and measures specified in the project description	The project activity instance shall implement improved cookstoves having minimum efficiency of 25% below which it shall not be eligible under this grouped project.	Thermal Efficiency test report/manufacture's specification shall be used for establishing minimum efficiency criteria in the first year of stove installation.
3.	Scale: Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit	i. The aggregate energy savings by a single project activity shall not exceed the equivalent of 180 GWh_{th}/year as mentioned in the methodology AMS II G: Energy efficiency measures in thermal applications of non-renewable biomass. ii. Each project activity instance that exceeds one percent of the capacity limit (i.e., 1.8 GWh_{th}) shall be identified. iii. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters.	i. Project Activity Instance Spreadsheet containing estimation of annual energy saving from each project activity instance (i.e., each project ICS). ii. Project Activity Instance ER Spreadsheet containing estimation of annual energy saving from each project activity instance (i.e., each project ICS). iii. The expected annual energy saving from each project activity instance (i.e., each project ICS) comes out to be in the range of 0.02 to 0.03 GWh_{th}/ ICS which is well below 1% (i.e., 1.8 GWh_{th}) of the capacity limit. Therefore, the project activity instances under this grouped

			project shall not be assigned to clusters.
4.	Project Location: Occur within one of the designated geographic areas specified in the project description	Each project activity instance shall be located within the geographic boundary of Bangladesh.	Self-declaration to demonstrate that project boundary is within the geographical boundary of Bangladesh. Project Activity Instance Database containing Geographical Coordinates of each household to which cookstoves are distributed.
5.	Start date	The start date of project activity instances shall be same as or later than the grouped project activity start date.	ICS distribution/registration data recorded in project database containing date of installation of each ICS.
6.	Additionality: Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	Each project activity instance shall demonstrate its compliance with a. Regulatory surplus The project should not be mandated by any law, statute, or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework. b. Positive List The inclusion of new project activity instances shall have to comply with positive list that is the ICS are installed/distributed at zero cost to end user and the project activity instance has no other source of revenue but GHG credits.	a. Regulatory Surplus There is no law, statute or government programme or policy in Bangladesh mandating the project activity nor is there any systematically enforced law, statute, or other regulatory framework for such projects. Hence this requirement is satisfied for all project activity instances to be included in the current grouped project. b. Positive List Self-declaration/ individual ICS registration form or any other documentary evidence mentioning that end users were not charged for the ICS.

			Same can be confirmed during verification.
7.	Target Population	End user for each project activity instance shall be households, with non-renewable biomass on inefficient traditional mud stove or three stone fire with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. Each ICS will be assigned a unique serial number with name of ICS user, address, GPS of household, stove model, distribution date, etc.	Database/ICS Registration form/ICS invoice if distribution has started or Self- declaration by project participant.
8.	Baseline Scenario: Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new project activity instances will be installed within Bangladesh only and the baseline scenario is the continued use of non-renewable wood fuel by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. This also conforms to the applied methodology.	Baseline scenario can be confirmed in Section 3.4 of this VCS-PD. Baseline scenario and geographical boundary will be confirmed from registration records cum consent deeds of individual ICS which will contain geographical coordinates, address as well as confirmation by end user (stove owner) stating that they were using traditional mud stove or three stone fire or conventional cook stove without a grate or a chimney before the installation of project stove.
9.	Double Counting	The Improved Cookstove	Project Activity Instance

		distributed under this project shall be uniquely identifiable based on the distribution records.	Database containing unique serial number of each ICS. Each ICS distributed will have corresponding end user details (i.e., name, geographical coordinates, address, unique serial number). The combination of this unique serial number of each ICS along with geographical address of the stove owner confirms the unique identity of each project device.
10.	Ownership: Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance	A default Beneficiary Agreement for end users including the provision that emission reductions generated by the project activity are owned by the Project Proponent will be provided for project activity instance.	Copy of registration form cum consent deed from end user (stove owner) to Project Proponent regarding emission reduction claims.
11.	Crediting period	Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4)	The start date of the instance will be the date, the stove began generating GHG emission reductions or removals. The end of the crediting period of instance will be the end date of project crediting period as mentioned in section 1.9.

		and new instances are eligible for crediting from the start of the next verification period.	
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1.5 Project Proponent

Organization name	C-Quest Capital BD Stoves 1 Pte Ltd
Contact person	Joel Sykes
Title	Director
Address	38 Beach Road, #29-11, South Beach Tower, Singapore 189767
Telephone	+1-202 247-7976
Email	cqc-operations@cquestcapital.com

1.6 Other Entities Involved in the Project

At present, C-Quest Capital BD Stoves 1 Pte Ltd is the sole entity involved in the project.

1.7 Ownership

The project ownership is with C-Quest Capital BD Stoves 1 Pte Ltd.

Further, registration certificate cum declaration signed by end user (stove owner) during the registration process includes the clause “Acknowledgement and Carbon Rights Waiver – My household hereby transfers and waives all and any rights in connection with the improved cookstove or project and greenhouse gas emission reductions or other environmental or social attributes generated by the installation and use of the improved cookstove to C-Quest Capital BD Stoves 1 Pte Ltd and we understand that C-Quest Capital BD Stoves 1 Pte Ltd may, at any time, transfer those rights to an affiliate without further notice or consent”.

This is in line with clause 5 of section 3.7.1. of VCS project standard version 04.4.

1.8 Project Start Date

01-July-2024³ (Date of commissioning of 1st ICS)

³ Date of the first stove installed under the grouped project activity. Due to operational constraints, PP couldn't start the implementation of the project as stated. It is now planned to start from 1st July 2024.

1.9 Project Crediting Period

01-July-2024 to 30-June-2034, ten years fixed crediting period.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☐ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☒ >1,000,000 tCO₂e/year

Project Scale	
Project	
Large project	X

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1 01-July-2024 to 30-June-2025	3,609,372
Year 2 01-July-2025 to 30-June-2026	9,659,351
Year 3 01-July-2026 to 30-June-2027	14,828,505
Year 4 01-July-2027 to 30-June-2028	13,897,458
Year 5 01-July-2028 to 30-June-2029	13,024,246
Year 6 01-July-2029 to 30-June-2030	12,205,306
Year 7 01-July-2030 to 30-June-2031	11,437,293
Year 8 01-July-2031 to 30-June-2032	10,717,066

Year 9 01-July-2032 to 30-June-2033	10,041,679
Year 10 01-July-2033 to 30-June-2034	9,408,363
Total estimated ERs	108,828,639
Total number of crediting years	10
Average annual ERs	10,882,863

1.11 Description of the Project Activity

The project involves distribution of fuel-efficient improved cookstoves (ICS) that replaces the existing inefficient, conventional open fire stoves. These improved cookstoves will burn more efficiently as compared to the baseline stoves and thus will save the amount of wood burnt in cooking purpose and emissions associated with it.

The ICS deployed will reach a specified efficiency of at least 25%. The technical description and technical specifications shall be detailed in respective Project activity instance. The type of ICS disseminated for usage under this Project Activity will be either portable or fixed ICS burning firewood. The project activity will generate greenhouse gas (GHG) emission reductions by avoiding CO₂ emissions as a function of the reduction in the amount of non- renewable biomass fuel consumption in the efficient project stoves as compared to baseline stoves.

The table below provides information on three examples of ICS - Baldwin 8" stove, Baldwin 10" stove and Dinabandhu stove, which may be distributed under this project activity. The ICS deployed under this project will substantially reduce fuel consumption and emissions for conducting cooking and water heating tasks in homes. Additional example of ICS to be deployed in this project activity will be provided during verification stage.



Baldwin portable stove

Table 1: Technical description of Baldwin stove (8")

No	Parameter	Value
1	Thermal Efficiency	41%
2	Dimensions	21.5 cm Diameter, 10 cm Depth
3	Material	- Skirt: Stainless Steel - Base: Mild Steel - Grate: Mild Steel - Cage: Galvanized Iron Wire
4	Life Span	10 years

Table 2: Technical description of Baldwin stove (10")

No	Parameter	Value
1	Thermal Efficiency	46%
2	Dimensions	24 cm Diameter, 10.5 cm Depth
3	Material	- Skirt: Stainless Steel - Base: Mild Steel - Grate: Mild Steel - Cage: Galvanized Iron Wire
4	Life Span	10 years

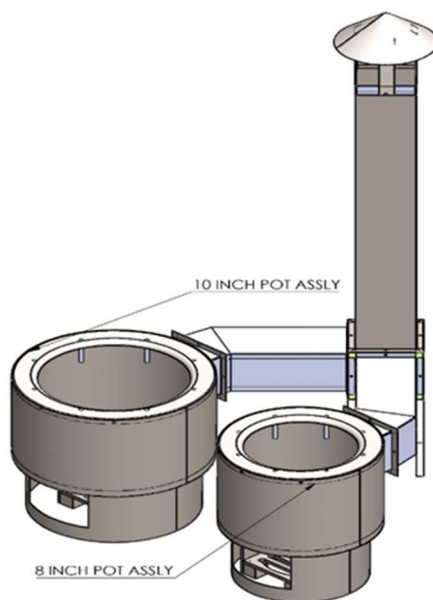


Table 3: Technical description of Baldwin Double Pot Stove

No	Parameter	Value
1	Thermal Efficiency	43%
2	Dimensions	24 cm Diameter, 10.5 cm Depth for 10" Stove 21.5 cm Diameter, 10 cm Depth for 8" Stove
3	Material	- Skirt: Stainless Steel - Base: Mild Steel - Grate: Mild Steel - Cage: Galvanized Iron Wire
4	Life Span	10 years

Dinabandhu stove is a type of in-situ fixed stove which will be constructed using locally available clay and dung along with some metal parts. The end users will be tutored for their use and regular upkeep, it is expected that the stoves will continue operating for several years with a little maintenance which is usually as simple as applying a paste of clay on the stove every now and then to arrest cracking and change of removable metal parts every few years. It can therefore be assumed that the life of the stoves will be more than 10 years thereby exceeding crediting period.



Schematic Diagram of 'Dinabandhu' Stove

Table 4: Technical specifications of Dinabandhu stove

No	Parameter	Value
1	Thermal Efficiency	34.6%
2	Dimensions	1.3 ft (H), 3 ft (L), 1.5 ft (W)
3	Material	Cow dung, clay, and a few metal parts such as chimney, grate etc.
4	Number of burners	2
5	Fuel	Woody Biomass
6	Life Span	10 years

Installation of the stoves will be taken up by trained masonry workers with help from households to get the stove dimensions right.



Dishari Green Stove

Table 5: Technical description of Dishari Green Stove

No	Parameter	Value
1	Thermal Efficiency	48.21%
2	Dimensions	20.32 cm Diameter, 22.86 cm Height
3	Material	- Combustion chamber: Stainless Steel - Outside body: Mild Steel - Handles: Metal wrapped with insulating material - Grate: Mild Steel (Perforated) - Fuel support: Wrought Iron
4	Life Span	10 years

Baseline cookstove

About 54.9% of rural population and 36.6% of urban population in Bangladesh still rely on woody biomass as their primary source of fuel⁴. As of 2019, only 23% of overall Bangladesh population has the access to clean fuels and technologies for cooking⁵. About 82 percent of Bangladesh's population relies on traditional mud stoves for cooking. Additionally, inefficient cooking produces greenhouse gas emissions, such as carbon dioxide and methane, and aerosols such as black carbon, contributing to adverse climate change. Women and young children spend a significant amount of time near cooking areas, inhaling pollutants from traditional stoves. Women who use

⁴ https://bbs.portal.gov.bd/sites/default/files/files/bbs.portal.gov.bd/page/b343a8b4_956b_45ca_872f_4cf9b2f1a6e0/2021-06-28-07-11-c1784c6fe700cf52ff934e8dd7cf9147.pdf (Bangladesh Bureau of Statistics: Bangladesh Sample Vital Statistics – 2020)

⁵ https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2021/Jun/SDG7_Tracking_Progress_2021.pdf

traditional stoves are particularly vulnerable to high levels of exposure to household air pollution (HAP): they are twice as likely to develop chronic obstructive pulmonary disease (COPD) than women who use cleaner stoves⁶.

It is a common and traditional practice in the project area to use a traditional mud stoves with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. This is also established by a baseline study carried out in the host country, Bangladesh⁷.

Data collection of ICS end-user

Project proponent must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number
- Retailer/distributor information

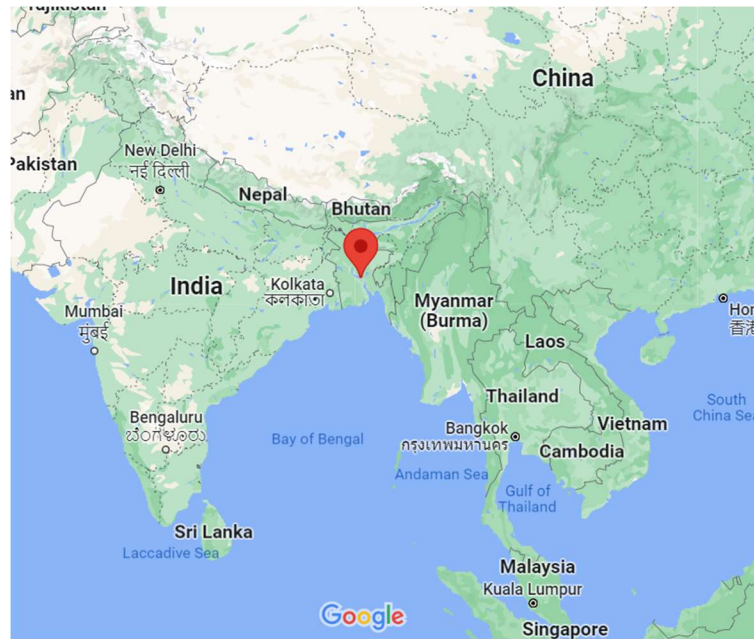
The information collected will be stored in the electronic database which will serve as project database for project monitoring and sampling purposes.

1.12 Project Location

The political boundary of Bangladesh will be the geographical boundary of the grouped project. The details of coordinates are being submitted separately as KML file.

⁶ <https://documents1.worldbank.org/curated/en/311561624871430872/pdf/Understanding-Market-Based-Solutions-and-Access-to-Finance-Options-for-Clean-Cooking-Technologies-in-Bangladesh.pdf> (2021 Report)

⁷ [Baseline Study Report – Bangladesh \(2023\)](#)



Grouped Project Location⁸



Map of project area

1.13 Conditions Prior to Project Initiation

⁸ Bangladesh Geographic coordinates - Latitude & longitude (geodatos.net)

As detailed in Section 1.11 and Section 3.4, the condition prior to project initiation is same as baseline scenario which is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by ICS in extremely inefficient traditional cookstoves. The pre-project devices are inefficient three-stone fires or conventional systems lacking improved combustion air supply mechanism and flue gas ventilation system, i.e., traditional stoves. Household air pollution (HAP) from cooking with solid fuels has become a leading cause of death and disability in many developing countries including Bangladesh⁹. Despite availability of ample evidence regarding adverse health effects of chronic exposure to biomass smoke generated indoors, it remains a widely used energy source in rural Bangladesh.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

A review is made on the following environmental laws and regulations in Bangladesh:

- Bangladesh Environment Conservation Act (1995)
- Bangladesh Renewable Energy Policy (2008)
- Sustainable Renewable Energy Development Authority SREDA Act (2012)
- Bangladesh National Women Policy (2011)
- National Energy policy (2004)
- Bangladesh Climate Change Strategy and Action Plan (2009)

There is neither any negative concern on improved cookstoves project nor any laws and regulations governing the use of improved cookstoves in Bangladesh households. The project is a voluntary effort by the project proponent.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The grouped project or any of the project activity instances included in it have not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

⁹ Household air pollution from cooking and risk of adverse health and birth outcomes in Bangladesh: a nationwide population-based study | Environmental Health | Full Text (biomedcentral.com)

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes ☒ No

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes ☒ No

The project has not sought or received another form of GHG-related environmental credit.

Supply Chain (Scope 3) Emissions

As per Section 3.23.9 of the VCS Standard, v4.4, the “producer(s) or retailer(s) of the impacted good or service are known but not involved in the project or do not have a website”. The Project Proponent will inform the manufacturer of ICS as well as the implementing partner in Bangladesh at the time of signing agreement for the goods and services provided by them, that ICS will be distributed in Bangladesh under this project activity. Also, PP will intimate them about the project registration under VERRA for the generation of VCUs.

PP has further apprised that the ownership of these credits lies exclusively with C-Quest Capital BD Stoves 1 Pte Ltd to avoid any potential risk of double claiming of Scope 3 emissions through notice publication. The same can be checked using the link below: <https://cquestcapital.com/latest/public-notice/>.

1.17 Sustainable Development Contributions

The details of the project activity’s contribution to the Sustainable Development Goals have been provided in the Table below:

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Contributions Over Project Lifetime

1	3.9	Project specific indicator: Reduction in PM _{2.5} emissions	Implemented activities to decrease	100% of the households receiving improved cookstoves under the project activity are expected to report reduced incidences of illness owing to smoke inhalation as well as reduced accidental burns.
2	4.3	Project specific indicator: Number of individuals who received any informal training	Increase	Contribute to increasing vocational and relevant skills of at least 50 local individuals (with a focus on targeting women and youth) by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to stove distribution and its monitoring under VCS
3	5.4	Project specific indicator: Proportion of women/female population reporting reduction in time spent in collecting fuelwood and cooking.	Implemented activities to decrease	100% of the primary fuelwood collectors/cooks in households receiving improved cookstoves under the project activity are expected to report reduction in time spent in collecting fuelwood and cooking activities, thereby freeing up time for more productive activities.
4	7.1	7.1.2 Proportion of population with primary reliance on clean fuels and technology	Increase	Increase access to clean cooking technology with project stoves installations in approximately 1,000,000 households under the project lifecycle
5	8.3	8.3.1 Proportion of informal employment in non-agriculture employment, by sex	Increase	Contribute to generation of employment in informal sector (total economy, agriculture, and non-agriculture) by contracting locals with a target employment of 50 employees for varying lengths of time over the project lifetime with a focus on hiring females. Employment comprises all persons of working age who, during a short reference period (minimum one week), or full time will be engaged in any activity to produce goods or provide services for pay or profit
6	13.0	Project Specific indicator: Reduction in emissions as compared to baseline scenario (open fire)	Increase	Contribute to GHG reduction through an estimated reduction of ~68 tCO ₂ e per stove for the ten-year period owing to the replacement of baseline stoves with project stoves.
7	15.2	15.2.1 Progress towards sustainable forest management by increasing above ground biomass in forests	Implemented activities to decrease	Contribute an estimated reduction of deforestation of ~5.05 tons of woody biomass, per stove, per year, from forests surrounding the communities and reducing pressure on forest reserves

1.18 Additional Information Relevant to the Project

Leakage Management

In line with the options available in the applied methodology, the Grouped Project applies a net to gross adjustment factor of 0.95 to account for any leakage resulting from it.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not Applicable

2 SAFEGUARDS

2.1 No Net Harm

The project will only bring positive impacts on environmental and socio-economic aspects as elaborated in Section 1.17. No potential negative environmental or socio-economic impacts have been identified for the project.

2.2 Local Stakeholder Consultation

Local Stakeholders Consultation has been undertaken at Project Level. Local Stakeholder Consultation are key components of the project operations. Project Proponent and Distribution Partners (on behalf of PP) undertake interactive training and demonstration sessions with community members who receive clean cookstoves. These sessions provide information on how to build/assemble, operate and area for the ICS as well as to engage with end users regarding queries that they may have in relation to the use of stoves.

The procedures or methods of conducting LSC and regular feedback mechanism

Local Stakeholders Consultation Meeting	The local stakeholder consultation (LSC) meeting for the VCS project – Installation of High Efficiency Wood Burning Cookstoves in Bangladesh was conducted on 6th May 2023 from 11:00 to 14:00 hours (Bangladesh Standard Time) both in offline and online mode.
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Mode of Invitation and identified stakeholders	The stakeholders were informed about the meeting across various platforms including advertisements in social media, personal invitation and mails sent to identified stakeholders such as NGOs, various government officials responsible for rural development, educationists, villagers, etc. Social media platforms such as Facebook, and WhatsApp were also used for broadcasting information about the webinar and inviting concerned stakeholders. The meeting was attended by nearly 42 stakeholders offline and 17 online modes. Among the participants were several personalities working in governing bodies, rural development, coastal livelihood impact and villagers who would be directly impacted by the Project Activity. (Invitation, Attendance records, and photographs attached in appendix for reference).
Response Mechanism	The stakeholders were requested to register their response through evaluation form (sample attached in appendix), or they could send their comments at the mail id provided.
Outcome of LSC meeting	The outcome of the LSC meeting more than encouraging. By the end of the meeting and even after, a number of stakeholders expressed their desire to connect with this programme as they truly felt that it could be a catalyst in bringing about household energy efficiency improvements in the rural and financially backward sections of society as well as preserve our pristine forests. None of the queries or comments raised by stakeholders (attached in Appendix), required any design change, albeit the proponents did realize that the suggestions put forth by stakeholders would go a long way in ensuring uptake and long-term use of the technology by the end users.

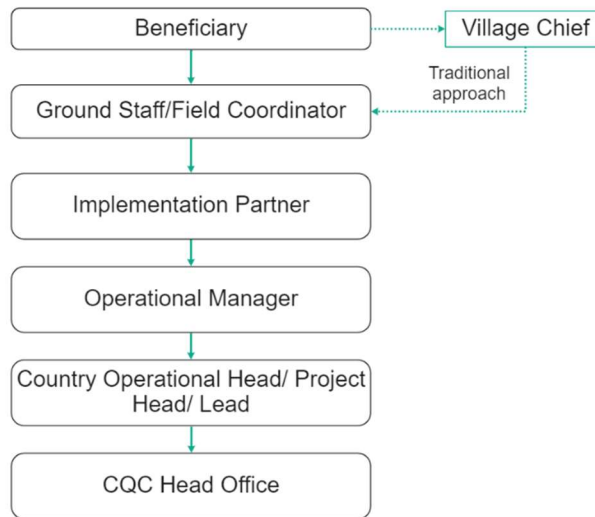
Further, PP follows three methods for continuous feedback and ongoing communications with stakeholders during project implementation. These are –

Annual follow-up visits to households – On-ground team/implementing partner makes at-least one visit to each registered household (ICS user) in a year. Usually, this follow-up visit is conducted within one month from the date of distribution/registration. The objective of this visit is to ensure that project cookstoves are being properly used and to remind women cooks on the project cookstoves, also to explain them about cookstove maintenance, best practices, and health benefit. This also addresses the grievances and provide maintenance to cookstoves as needed.

Call over helpline number - End Users/beneficiaries can report their grievances and complaints regarding stove parts or any other issues via calling on the helpline number which has been engraved on the ICS serial number plate or provided them in the pamphlet. The CQC ground team/implementing partner takes immediate actions to address the grievances.

Spot check visit – The CQC ground team makes visit to registered households on sample basis. One of the main objectives of this visit is to ensure that the implementation partner is addressing the concerns from the beneficiaries in adequate way and the end-users are trained on operation and maintenance of cookstoves including making the end-users aware about the benefits associated with the cookstoves.

Grievance Redressal Mechanism - PP follows a proper Feedback and Grievance Redress Policy and Procedure, which ensures that project-affected communities and individual stakeholders' grievances are properly prioritized and addressed. These measures are taken to enhance PP's accountability and transparency and to support the project initiatives that can help the communities to identify adverse effects to them, or their environment which had not previously been identified and mitigated by PP.



Flow of information/complaint/grievances from stakeholder to PP

The Grievance Policy and Procedure will be accessible as a phone number of the PP/implementing partner office in Bangladesh which will be made available on the brochure provided to each household.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

No comments were received for this project during the public commenting period from 20 April 2023 to 20 May 2023.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

- VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1¹⁰
- Tool 30: “Calculation of the fraction of non-renewable biomass” (version 4.0¹¹)
- Standard: “Sampling and Surveys for CDM project activities and programmes of activities” (version 09.0¹²)

3.2 Applicability of Methodology

Applicability criterion	How the project complies
Project activities shall be implemented in domestic premises or in community-based kitchen	The proposed project involves deployment of ICS in domestic premises only.
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot;	Five different models of ICS planned to be installed under this project are Baldwin 8”, Baldwin 10”, Dishari Green Stove, Baldwin Double Pot, and Dinabandhu stove with efficiency of 41%, 46%, 48.21%, 43% and 34.6% respectively as per the manufacturer’s specifications/efficiency test report.

¹⁰ <https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

¹¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf>

¹² https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics;	As the use of non-renewable biomass in Bangladesh since 31st December 1989 has been established, the present project activity instance is not required to demonstrate the same.
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips). it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Not applicable. The ICS is introduced as energy efficiency measure to replace baseline stoves and reduce the use of non-renewable biomass for combustion.
Both 'Projects' and 'Large Projects' can use this methodology. Each cookstove shall result in annual energy saving of less than or equal to 1% of the capacity limit indicated in the methodology.	The proposed project is expected to generate 10,882,863 tonnes CO ₂ per year which is greater than 300,000 tonnes CO ₂ per year and thus qualifies as a "large project" as per section 3.9.1 of the VCS Standard version 4.4.
The PD shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	Each ICS in this project will be identified by a unique combination of customer name and geographical location, as well as an alphanumeric serial number. The serial number will be a unique number which will allow for a clear distinction between the stoves. No individual serial number can be repeated within the project, thus ensuring that each stove is counted only once in the proposed project.
The PD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale	The stove manufacturers/wholesale providers/end users shall sign an undertaking stating clearly that the PP or an entity authorized by it shall be the sole owner of the VCU's arising from the project.

providers or others claim credit for emission reductions from the project devices.	
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Evidence that the non-renewable biomass has been in use since 1989

Non-renewable biomass has been in use since 31 December 1989 as evidenced by various FAO statistical data. The Global Forest Resources Assessment Country Report (2020)¹³ (FAO) indicates that between years 1990 and 2020, Bangladesh lost 37,000 hectares of forest during the period. This amounts to a deforestation rate of 1.93%. Forest area has decreased from 1,920,000 ha in year 1990 to 1,883,000 ha in year 2020.

Table: Extent of forest area between 1990-2020

Country	Extent of Forest area (1000 hectares)			
	1990	2000	2010	2020
Bangladesh	1920	1920	1888	1883

Thus, it can be concluded that losses have been reported in forest areas from before 1989 thereby hinting that fuelwood used across Bangladesh comes from non-renewable sources.

3.3 Project Boundary

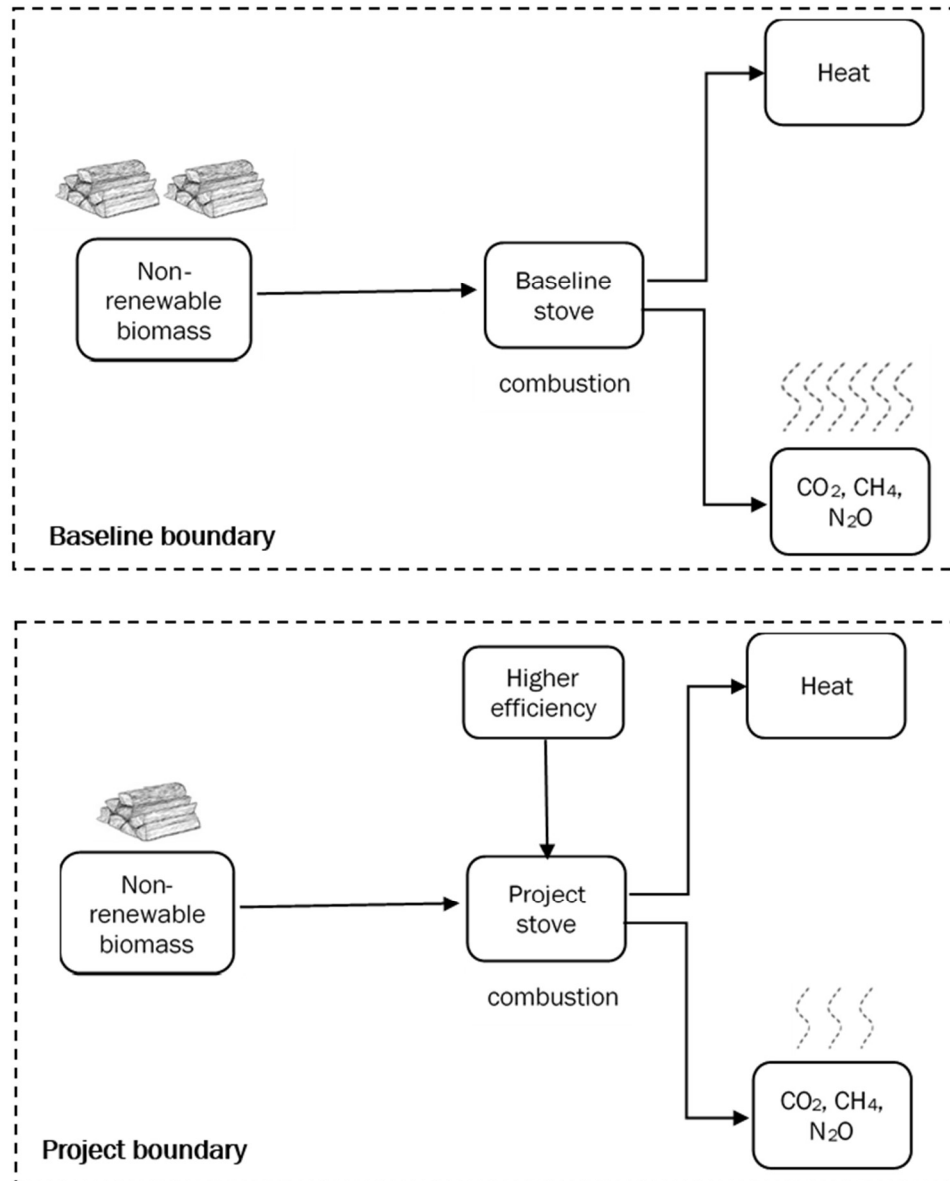
The project boundary for this project is the geographical boundary of Bangladesh. Refer section 1.12 for more details .

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable biomass/Fossil fuel	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No other source identified
Project	Emission from use of non-renewable biomass	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source

¹³ FAO Global Forest Resources Assessment 2020, Global Tables, Table A1
(<https://www.fao.org/3/ca9825en/ca9825en.pdf>)

Source	Gas	Included?	Justification/Explanation
	Other	No	No other source identified

A representation of the baseline boundary and project boundary are given as below.



3.4 Baseline Scenario

The baseline scenario is the continued use of non-renewable wood fuel (firewood) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

About 54.9% of rural population and 36.6% of urban population in Bangladesh still rely on woody biomass as their primary source of fuel¹⁴. As of 2019, only 23% of overall Bangladesh population has the access to clean fuels and technologies for cooking¹⁵. About 82 percent of Bangladesh's population relies on traditional mud stoves for cooking. Additionally, inefficient cooking produces greenhouse gas emissions, such as carbon dioxide and methane, and aerosols such as black carbon, contributing to adverse climate change. Women and young children spend a significant amount of time near cooking areas, inhaling pollutants from traditional stoves. Women who use traditional stoves are particularly vulnerable to high levels of exposure to household air pollution (HAP): they are twice as likely to develop chronic obstructive pulmonary disease (COPD) than women who use cleaner stoves¹⁶.

For many years, locals in Northern Bangladesh uses a traditional mud stove, the chulha, in a badly ventilated kitchen. The discomfort of respiratory trouble and persistent coughs had become so routine that no one blinked an eye. Every year, indoor air pollution contributes to 49,000 premature deaths in Bangladesh¹⁷.

It is a common and traditional practice in the project area to use a traditional mud stoves with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. This is also established by a baseline study carried out in the host country, Bangladesh¹⁸ and the same report has been shared with VVB. There is no other type of stove or technology being used or observed in the project area. Hence, the efficiency for chulha or conventional stoves stands at 0.1 or 10% as per the applied methodology and the same is considered in the baseline estimations.

3.5 Additionality

The methodology uses activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute, or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework.

¹⁴ https://bbs.portal.gov.bd/sites/default/files/files/bbs.portal.gov.bd/page/b343a8b4_956b_45ca_872f_4cf9b2f1a6e0/2021-06-28-07-11-c1784c6fe700cf52ff934e8dd7cf9147.pdf (Bangladesh Bureau of Statistics: Bangladesh Sample Vital Statistics – 2020)

¹⁵ https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2021/Jun/SDG7_Tracking_Progress_2021.pdf

¹⁶ <https://documents1.worldbank.org/curated/en/311561624871430872/pdf/Understanding-Market-Based-Solutions-and-Access-to-Finance-Options-for-Clean-Cooking-Technologies-in-Bangladesh.pdf> (2021 Report)

¹⁷ [In Bangladesh, women's right to clean air starts in the kitchen \(unep.org\)](#) (2019)

¹⁸ [Baseline Study Report – Bangladesh \(2023\)](#)

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by CQC.

Step 2: Positive List

If the project installs the ICS at zero cost to the household and has no other source of revenue other than the sale of GHG credits and it is not implemented as part of any government scheme or supported by multilateral funds, then the project is automatically additional.

Conclusion: As the project fulfils the conditions above, it is deemed additional.

Step 3: Project Method

For any project activity where stoves are not provided at zero cost to the end-user or has any other source of revenues other than the sale of GHG credits, then the project activity shall apply investment analysis method set out in the CDM Tool for the Demonstration and Assessment of Additionality included in AMS-II.G to determine that the proposed project activity is either: 1) not the most economically or financially attractive, or 2) not economically or financially feasible.

This step is not applicable under project activity because the stoves are provided at zero cost to the end-users.

3.6 Methodology Deviations

The grouped project has not applied any methodology deviation.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The methodology (Methodology for Installation of High Efficiency Firewood Cookstoves v.1.1) does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.2 Project Emissions

The methodology (Methodology for Installation of High Efficiency Firewood Cookstoves v.1.1) does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.3 Leakage

Leakage shall be considered as default 0.95 in accordance with methodology (Methodology for Installation of High Efficiency Firewood Cookstoves v.1.1).

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}$$

Where:

i	=	Indices for the situation where more than one type/model of improved cookstove is introduced to replace traditional stove or three-stone fire
j	=	Indices for the situation where there is more than one batch of improved cookstove of type i
ER_y	=	Emission reductions during year y in t CO ₂ e
$ER_{y,i,j}$	=	Emission reductions by improved cookstove of type i and batch j during year y in t CO ₂ e

$$ER_{y,i,j} = B_{y,savings,i,j} \times f_{NRB,y} \times NCV_{wood\ fuel} \times (EF_{wf,CO2} + EF_{wf,non\ CO2}) \times N_{y,i,j} \times 0.95 \quad \text{Equation (2)}$$

Where:

$B_{y,savings,i,j}$	=	Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y
$f_{NRB,y}$	=	Fraction of woody biomass that can be established as non-renewable biomass (fNRB) ¹⁹
$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne) ²⁰
$EF_{wf,CO2}$	=	CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO ₂ /TJ) ²¹

¹⁹ Calculated as per CDM Tool 30 version 4.0

²⁰ AMS II.G. Version 13

²¹ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Chapter 2 Stationary Combustion

$EF_{wf, non\ CO2}$	= Non-CO2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO ₂ /TJ) ²²
$N_{y,i,j}$	= Number of improved cook stoves of type i and batch j operating during year y
0.95	= Discount factor to account for leakage

The quantify of woody biomass saved due to implementation of improved cookstoves to be estimated using equation below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right) \quad \text{Equation (4)}$$

Where:

η_{old}	= Efficiency of baseline cookstove
$\eta_{new,y,i,j}$	= Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y . Alternatively, efficiency may be determined using Equation 4.
$B_{y=1,new,i,j,survey}$	= Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j , determined in the first year of the implementation of the project through a sample survey.

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad \text{Equation (5)}$$

where

η_p	= Efficiency of project stove (fraction) at the start of project activity.
$(DF_n)^{y-1}$	= Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.
0.94	= Adjustment factor to account for uncertainty related to project cookstove efficiency test.

For ex-ante calculation purpose, the assumption below is applied.

- 1) The project will install up to 1,600,000 ICS of which 600,000 will be Baldwin stove (8 inch), 400,000 will be Baldwin stove (10 inch), 200,000 will be Dishari Green Stove, 200,000 will be Baldwin Double Pot and 200,000 will be Dinabandhu stove.
- 2) Estimated implementation plan for portable and fixed ICS are assumed as below:

²² 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Chapter 2 Stationary Combustion

- Portable Stove- (20% in 1st year, 40% in 2nd year, and remaining 40% in 3rd year)
- Fixed Stove- (30% in 1st year, 40% in 2nd year, and remaining 30% in 3rd year)

3) The life span of ICS is 10 years.

4) $B_{y=1,new,i,j,survey}$ is assumed as below:

- Baldwin 8" Stove: 1.75 tonnes/device/yr
- Baldwin 10" Stove: 1.75 tonnes/device/yr
- Dishari Green Stove: 1.28 tonnes/device/yr
- Baldwin Double Pot Stove: 2.19 tonnes/device/yr
- Dinabandhu Stove: 2.15 tonnes/device/yr

5) Fraction of non-renewable biomass (f_{NRB}) calculation for Bangladesh

Summary of the equations applied to estimate f_{NRB} according to the latest version of the CDM Tool 30 (EB 115 Annex 22 version 4.0 2022).

The fraction of harvested biomass that can be established as non-renewable biomass is the f_{NRB} . This fraction ranges between 0 and 1, where 0 indicates that 100% of harvested biomass is renewable (0% is non-renewable) and 1 indicates that 100% is non-renewable. The following equation is used to calculate f_{NRB}:

$$f_{NRB} = \frac{NRB}{NRB + RB}$$

Where:

f_{NRB} = Fraction of non-renewable biomass

NRB = Non-renewable biomass (t/yr)

RB = Renewable biomass (t/yr)

The quantity of non-renewable biomass (NRB) is the difference between the annual biomass consumption (H) and the quantity of renewable biomass (RB), calculated by the following equation:

$$NRB = H - RB$$

Where:

H = Total woody biomass consumption in the absence of project activity (t/yr)

Renewable biomass is the product of the mean annual increment of woody biomass growth and the total extent of the forest and other wooded areas of the country where wood extraction is not prohibited or geographically remote, calculated by the following equation:

$$RB = \sum(MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum(MAI_{other,i} \times (F_{other,i} - P_{other,i}))$$

Where:

MAI = Mean annual increment of woody biomass (t/ha/year)

F_{forest} = Extent of forest in the relevant period (ha)

F_{other} = Extent of other land in the relevant period (ha)

P_{forest} = Extent of non-accessible area within forest areas (ha)

P_{other} = Extent of non-accessible area within other land areas (ha)

i = Sub-category i of forest areas and other land areas

The total woody biomass consumption in a country comprises its domestic and non-domestic energy and non-domestic non-energy consumption. For Bangladesh, this total is estimated using consumption data sourced from the FAO Forest Products Databases and World Bank population statistics. The reported volumes of fuelwood consumed in cubic meters are converted to metric tonnes using the FAO default conversion factor. Finally, consumed volumes of commercial forest products – including industrial roundwood, sawn wood, veneer sheets and wood-based panels are converted to metric tonnes using a biomass conversion and expansion factor provided for the region.

Table: Total 2021 population, woody biomass consumption, and per capita consumption estimates for Bangladesh:

Variable	Value	Source
Population		
Total	166,303,494	The World Bank ²³
Urban	64,768,559	The World Bank
Rural	101,534,935	The World Bank
Domestic woody biomass consumption		
Wood fuel (t/yr)	11,483,601	Calculated
Domestic per capita consumption rates		
Wood fuel (m3/capita/yr)	0.156	FAO ²⁴
Total Non-domestic (energy) wood consumption		
Wood charcoal (t/yr)	1,330,428	Calculated
Total Non-domestic (non-energy) wood consumption		
Total non-domestic (non-energy) (t/yr)	1,086,571	Calculated
Industrial roundwood (m3/yr)	498,910	Calculated
Sawn wood (m3/yr)	332,607	Calculated

²³ The World Bank. World Bank Open Data. <https://data.worldbank.org/> (2022).

²⁴ FAO Statistics. Forest Products 2019. <https://www.fao.org/3/cb3795m/cb3795m.pdf> (2021).

Veneer sheets (m3/yr)	6,120	Calculated
Wood-based panels (m3/yr)	31,619	Calculated
Non-domestic per capita consumption per category		
Wood charcoal (t/capita/yr)	0,002	FAO
Industrial roundwood (m3/capita/year)	0,003	FAO
Sawn wood (m3/capita/yr)	0,002	FAO
Veneer sheets (m3/capita/yr)	<0,001	FAO
Wood-based panels (m3/capita/yr)	<0,001	FAO
Conversion factors		
Wood density (t/m3)	0.725	FAO ²⁵
Biomass conversion and expansion factor (t/m3)	1.250	FAO ²⁶
Wood to charcoal conversion factor for urban population	4.000	CDM
Total woody biomass consumption (t/yr)	13,900,600	Calculated

Table: Total, protected and remote forest cover extent, mean annual increment (MAI), and renewable biomass (RB) by Global Ecological Zone for Bangladesh.

Ecological Zone	Total forest cover (ha)	Protected cover (ha)	Remote cover (ha)	MAI (t/ha/yr)	Renewable biomass (t/yr)
Tropical moist forest	353,610	37,924	139,505	0.89	157,616
Tropical rainforest	915,926	63,224	449,143	2.57	1,035,188
Total	1,269,536	101,148	588,648	-	1,192,804

The difference between woody biomass consumption (13,900,600 t/yr) and renewable biomass (1,192,804 t/yr) is considered to be non-renewable (CDM Tool 30, Equation 2). Non-renewable biomass utilization in Bangladesh is, therefore, estimated to be 1,192,804 t/yr.

²⁵ FAO. The charcoal transition. Greening the charcoal value chain to mitigate climate change and improve local livelihoods. (2017).

²⁶ FAO. Global Forest Resources Assessment 2020: Main report. Rome <https://doi.org/10.4060/ca9825en> (2020).

The resulting non-renewable biomass (NRB) was compared to the top-down product of average above-ground biomass (62.82 t/ha)²⁷ and the most recent annual deforestation data (21,475 ha/yr)²⁸. Non-renewable biomass (NRB) was found to be 12,707,796 t/yr, which is greater than the cross-check results based on deforestation (1,349,060 t/yr). It is expected that the cross-check estimate of biomass loss from deforestation could underestimate the extent of non-renewable biomass as it only considers the average biomass stocks from completely deforested areas. Unsustainable wood harvesting resulting in forest degradation and substantial reductions in biomass stocks, but not complete deforestation is not considered in the cross-check. Consequently, the estimate of NRB is considered applicable for this study given that the guidance provided by CDM Tool 30 v4.0 2022²⁹ was followed.

Variable	Value	Source
Woody biomass consumption (t/yr)	13,900,600	The World Bank, FAO
Renewable biomass (t/yr)	1,192,804	Calculated
Non-renewable biomass (t/yr)	12,707,796	Calculated (Equation 2 of CDM tool 30)
Fraction of non-renewable biomass	0.91	Calculated (Equation 1 of CDM Tool 30)

The fraction of non-renewable biomass is the quotient of the non-renewable and the total biomass (CDM Tool 30, Equation 1). The fraction of non-renewable biomass for Bangladesh is, therefore, estimated to be 0.91. The fNRB report and calculation spreadsheet has been shared with VVB.

6) Annual stove loss rate is assumed as 5%.

Determination of number of ICS operating during year y

For Baldwin 8" Stove (portable),

$$N_{y,i,j} = 600,000 \times [1 - (y-1) \times 5\%]$$

Example of calculation:

If y = 1,

²⁷ Food and Agriculture Organization of the United Nations. Global forest resources assessments - country profiles. <https://www.fao.org/forest-resources-assessment/fra-2020/country-reports/en/> (2020).

²⁸ Mongabay. Rainforest information. <https://rainforests.mongabay.com/> (2020).

²⁹ CDM. Methodological tool 30: Calculation of the fraction of non-renewable biomass. Version 04.0. <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf> (2022).

$$N_{y,i,j} = 20\% \text{ of } 600,000 \times [1 - (1-1) \times 5\%]$$

$$= 120,000 \times 100\%$$

$$= 120,000$$

For, $y=2$

$$N_{y,i,j} = 20\% \text{ of } 600,000 \times [1 - (2-1) \times 5\%] + 40\% \text{ of } 600,000$$

$$= 120,000 \times 95\% + 240,000$$

$$= 354,000$$

Similarly, the number of ICS operating during year y for all ICS models are shown below:

	Baldwin stove 8 inch	Baldwin stove 10 inch	Dishari Green Stove	Baldwin Double Pot	Dinabandhu stove
Year (y)	$N_{y,i,j}$	$N_{y,i,j}$	$N_{y,i,j}$	$N_{y,i,j}$	$N_{y,i,j}$
01-July-2024 to 30-June-2025	120,000	80,000	40,000	60,000	60,000
01-July-2025 to 30-June-2026	354,000	236,000	118,000	137,000	137,000
01-July-2026 to 30-June-2027	576,300	384,200	192,100	190,150	190,150
01-July-2027 to 30-June-2028	547,485	364,990	182,495	180,643	180,643
01-July-2028 to 30-June-2029	520,111	346,741	173,370	171,610	171,610
01-July-2029 to 30-June-2030	494,105	329,403	164,702	163,030	163,030
01-July-2030 to 30-June-2031	469,400	312,933	156,467	154,878	154,878
01-July-2031 to 30-June-2032	445,930	297,287	148,643	147,134	147,134
01-July-2032 to 30-June-2033	423,633	282,422	141,211	139,778	139,778
01-July-2033 to 30-June-2034	402,452	268,301	134,151	132,789	132,789

Determination of efficiency of ICS during year y

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where

η_p for different ICS models are shown below:

Baldwin 8" Stove: 41%

Baldwin 10" Stove: 46%

Dishari Green Stove: 48.21%

Baldwin Double Pot Stove: 43%

Dinabandhu Stove: 34.60%

$$DF_n = 0.99$$

Example of calculation:

for Baldwin 8" Stove, if $y = 2$

$$\begin{aligned}\eta_{new,y,i,j} &= 41\% \times (0.99)^{2-1} \times 0.94 \\ &= 38.15\%\end{aligned}$$

Similarly, the efficiency of ICS during year y for all ICS models are shown below:

	Baldwin stove 8 inch	Baldwin stove 10 inch	Dishari Green Stove	Baldwin Double Pot	Dinabandhu stove
Year (y)	$\eta_{new,y,i,j}$	$\eta_{new,y,i,j}$	$\eta_{new,y,i,j}$	$\eta_{new,y,i,j}$	$\eta_{new,y,i,j}$
01-July-2024 to 30-June-2025	38.54%	43.24%	45.32%	40.42%	32.52%
01-July-2025 to 30-June-2026	38.15%	42.81%	44.86%	40.02%	32.20%
01-July-2026 to 30-June-2027	37.77%	42.38%	44.42%	39.62%	31.88%
01-July-2027 to 30-June-2028	37.40%	41.96%	43.97%	39.22%	31.56%
01-July-2028 to 30-June-2029	37.02%	41.54%	43.53%	38.83%	31.24%
01-July-2029 to 30-June-2030	36.65%	41.12%	43.10%	38.44%	30.93%
01-July-2030 to 30-June-2031	36.28%	40.71%	42.67%	38.05%	30.62%
01-July-2031 to 30-June-2032	35.92%	40.30%	42.24%	37.67%	30.31%
01-July-2032 to 30-June-2033	35.56%	39.90%	41.82%	37.30%	30.01%
01-July-2033 to 30-June-2034	35.21%	39.50%	41.40%	36.92%	29.71%

Determination of quantity of woody biomass that is saved in tonnes per ICS during year y

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right)$$

Example of calculation:

For Baldwin 8" Stove, If $y = 2$,

$$\begin{aligned}B_{y,savings,i,j} &= 1.75 \times [(0.3815/0.1) - 1] \\ &= 4.93 \text{ tonnes}\end{aligned}$$

Similarly, the quantity of woody biomass that is saved in tonnes per ICS during year y for all ICS models are shown below:

	Baldwin stove 8 inch	Baldwin stove 10 inch	Dishari Green Stove	Baldwin Double Pot	Dinabandhu stove
Year (y)	$B_{y,savings,i,j}$	$B_{y,savings,i,j}$	$B_{y,savings,i,j}$	$B_{y,savings,i,j}$	$B_{y,savings,i,j}$
01-July-2024 to 30-June-2025	5.0002	5.8236	4.5118	6.6620	4.8341
01-July-2025 to 30-June-2026	4.9327	5.7479	4.4539	6.5735	4.7643
01-July-2026 to 30-June-2027	4.8658	5.6729	4.3966	6.4858	4.6952
01-July-2027 to 30-June-2028	4.7997	5.5986	4.3398	6.3991	4.6268
01-July-2028 to 30-June-2029	4.7341	5.5251	4.2837	6.3132	4.5590
01-July-2029 to 30-June-2030	4.6693	5.4524	4.2281	6.2281	4.4920
01-July-2030 to 30-June-2031	4.6051	5.3803	4.1730	6.1440	4.4256
01-July-2031 to 30-June-2032	4.5415	5.3090	4.1185	6.0606	4.3599
01-July-2032 to 30-June-2033	4.4786	5.2384	4.0645	5.9781	4.2948
01-July-2033 to 30-June-2034	4.4163	5.1685	4.0111	5.8964	4.2304

Determination of emission reductions by ICS of batch 1 during year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where

$$NCV_{wood\ fuel} = 0.0156 \text{ TJ/tonne}$$

$$f_{NRB,y} = 0.91$$

$$EF_{wf,CO} + EF_{wf,non\ CO_2} = 112 + 26.23 = 138.23 \text{ tCO}_2/\text{TJ}$$

Example of calculation:

For Baldwin 8" stove, If y=2,

$$\begin{aligned} ER_{y,i,j} &= 4.93 \times 0.0156 \times 0.91 \times 138.23 \times 354,000 \times 0.95 \\ &= 3,255,207 \text{ tCO}_2 \end{aligned}$$

Similarly, emission reductions by ICS of batch 1 during year y for all ICS models are shown below:

	Baldwin stove 8 inch	Baldwin stove 10 inch	Dishari Green Stove	Baldwin Double Pot	Dinabandhu stove	Total (Round down)
Year (y)	$ER_{y,i,j}$	$ER_{y,i,j}$	$ER_{y,i,j}$	$ER_{y,i,j}$	$ER_{y,i,j}$	tCO ₂
01-July-2024 to 30-June-2025	1,118,565	868,514	336,435	745,155	540,703	3,609,372
01-July-2025 to 30-June-2026	3,255,207	2,528,788	979,749	1,678,829	1,216,778	9,659,351

01-July-2026 to 30-June-2027	5,227,551	4,063,066	1,574,473	2,299,077	1,664,339	14,828,505
01-July-2027 to 30-June-2028	4,898,630	3,809,392	1,476,446	2,154,907	1,558,083	13,897,458
01-July-2028 to 30-June-2029	4,590,175	3,571,409	1,384,469	2,019,684	1,458,511	13,024,246
01-July-2029 to 30-June-2030	4,300,921	3,348,151	1,298,170	1,892,857	1,365,207	12,205,306
01-July-2030 to 30-June-2031	4,029,686	3,138,716	1,217,203	1,773,909	1,277,781	11,437,293
01-July-2031 to 30-June-2032	3,775,355	2,942,252	1,141,239	1,662,354	1,195,866	10,717,066
01-July-2032 to 30-June-2033	3,536,885	2,757,964	1,069,973	1,557,738	1,119,119	10,041,679
01-July-2033 to 30-June-2034	3,313,296	2,585,103	1,003,115	1,459,631	1,047,219	9,408,363

Year	Estimated baseline emissions or removals (tCO2e)	Estimated project emissions or removals (tCO2e)	Estimated leakage emissions(tCO2e)	Estimated net GHG emission reductions or removals (tCO2e)
Year 2024-25 01-July-2024 to 30-June-2025	-	-	-	3,609,372
Year 2025-26 01-July-2025 to 30-June-2026	-	-	-	9,659,351
Year 2026-27 01-July-2026 to 30-June-2027	-	-	-	14,828,505
Year 2027-28 01-July-2027 to 30-June-2028	-	-	-	13,897,458
Year 2028-29 01-July-2028 to 30-June-2029	-	-	-	13,024,246
Year 2029-30 01-July-2029 to 30-June-2030	-	-	-	12,205,306
Year 2030-31 01-July-2030 to 30-June-2031	-	-	-	11,437,293
Year 2031-32 01-July-2031 to 30-June-2032	-	-	-	10,717,066
Year 2032-33 01-July-2032 to	-	-	-	10,041,679

30-June-2033				
Year 2033-34 01-July-2033 to 30-June-2034	-	-	-	9,408,363
Total	-	-	-	108,828,639

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	Official statistics or reports or peer-reviewed literature (UN Data & FAO reports)
Value applied	0.91
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante. C4 EcoSolutions (Pty) Ltd was appointed as third party to study and derive the f_{NRB} value for Bangladesh.
Purpose of Data	Calculation of emission reductions
Comments	The report of f_{NRB} has been shared to VVB.

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced

Source of data	Methodology default value; 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction ³⁰
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	EF_{wf,CO_2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	Methodology default value; 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion ³¹
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	$EF_{wf,non\ CO_2}$
Data unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario

³⁰ https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_1_Ch1_Introduction.pdf

³¹ https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf

Source of data	Methodology Default value; 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion ³²
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity.
Source of data	Manufacturer's specification/ WBT efficiency test report
Value applied	Baldwin 8" Stove: 0.41 Baldwin 10" Stove: 0.46 Dishari Green Stove: 0.4821 Baldwin Double Pot Stove: 0.43 Dinabandhu Stove: 0.3460
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
------------------	-------------

³² https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf

Data unit	Number
Description	Number of project devices of type i and batch j operating during year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities (v9.0).
Frequency of monitoring/recording	At least once every two years
Value applied	For ex-ante emission reduction calculation, it is assumed that the project will install 600,000 Baldwin 8" stoves, 400,000 Baldwin 10" stoves, 200,000 Dishari Green Stoves, 200,000 Baldwin Double Pot stoves, and 200,000 Dinabandhu stoves in this project activity.
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	The unique reference number of each stove shall be logged in the monitoring database showing the total number of stoves. Data from the sampling plan will be collected in each monitoring period by trained project staff and applied in the emissions reduction calculations. Internal cross-checks by the Project Proponent will be undertaken as Quality Check.
Purpose of data	Calculation of emission reductions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value
Comments	-

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline stove
Source of data	Methodological default value of 0.1

Description of measurement methods and procedures to be applied	A default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.
Frequency of monitoring/recording	Fixed throughout the crediting period for each individual household at the time of project implementation.
Value applied	0.1
Monitoring equipment	Not Applicable
QA/QC procedures to be applied	Not Applicable
Purpose of Data	Calculation of emission reductions
Comments	A default value of 0.1 shall be used and is going to remain fixed throughout the crediting period, as the project intends to target only those households that are using the traditional mud stove or three stone fire or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.

Data / Parameter	$\eta_{new,y,i,j}$																								
Data unit	Fraction																								
Description	Efficiency of the device of each type i and batch j implemented as part of the project activity																								
Source of data	Calculation																								
Description of measurement methods and procedures to be applied	To adopt equation 5 given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 4 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”																								
Frequency of monitoring/recording	Once at the time of project stove installation																								
Value applied	For ex-ante calculation, the value below is applied. <table><tr><th>Year (y)</th><th>Baldwin stove 8 inch (%)</th><th>Baldwin stove 10 inch (%)</th><th>Dishari Green Stove (%)</th><th>Baldwin Double Pot (%)</th><th>Dinabandhu stove (%)</th></tr><tr><td>1</td><td>38.54</td><td>43.24</td><td>45.32</td><td>40.42</td><td>32.52</td></tr><tr><td>2</td><td>38.15</td><td>42.81</td><td>44.86</td><td>40.02</td><td>32.20</td></tr><tr><td>3</td><td>37.77</td><td>42.38</td><td>44.42</td><td>39.62</td><td>31.88</td></tr></table>	Year (y)	Baldwin stove 8 inch (%)	Baldwin stove 10 inch (%)	Dishari Green Stove (%)	Baldwin Double Pot (%)	Dinabandhu stove (%)	1	38.54	43.24	45.32	40.42	32.52	2	38.15	42.81	44.86	40.02	32.20	3	37.77	42.38	44.42	39.62	31.88
Year (y)	Baldwin stove 8 inch (%)	Baldwin stove 10 inch (%)	Dishari Green Stove (%)	Baldwin Double Pot (%)	Dinabandhu stove (%)																				
1	38.54	43.24	45.32	40.42	32.52																				
2	38.15	42.81	44.86	40.02	32.20																				
3	37.77	42.38	44.42	39.62	31.88																				

	4	37.40	41.96	43.97	39.22	31.56
	5	37.02	41.54	43.53	38.83	31.24
	6	36.65	41.12	43.10	38.44	30.93
	7	36.28	40.71	42.67	38.05	30.62
	8	35.92	40.30	42.24	37.67	30.31
	9	35.56	39.90	41.82	37.30	30.01
	10	35.21	39.50	41.40	36.92	29.71
Monitoring equipment	Not Applicable as it is calculated.					
QA/QC procedures to be applied	Not Applicable as it is calculated.					
Purpose of data	Calculation of emission reductions					
Calculation method	Calculation to be performed using equation below: $\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$					
Comments	No comments					

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes
Description	Quantity of woody biomass used by project devices in tonnes per device of type i
Source of data	Monitoring survey (In the first year of introduction of ICS)
Description of measurement methods and procedures to be applied	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for CDM project activities and programme of activities (v9.0). Determined in the first year of the introduction of the devices
Frequency of monitoring/recording	Determine in the first year of project implementation
Value applied	For ex-ante calculation, the value is assumed as below: Baldwin 8" Stove: 1.75 tonnes/device/yr Baldwin 10" Stove: 1.75 tonnes/device/yr Dishari Green Stove: 1.28 tonnes/device/yr Baldwin Double Pot Stove: 2.19 tonnes/device/yr

	Dinabandhu Stove: 2.15 tonnes/device/yr
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	Survey to be conducted as per requirements of the methodology. Monitoring equipment will be calibrated as per the manufacturer 's guidelines.
Purpose of data	Calculation of emission reductions
Calculation method	Mean woody biomass consumption of sample stoves to be applied across entire population using a particular model of stove.
Comments	-

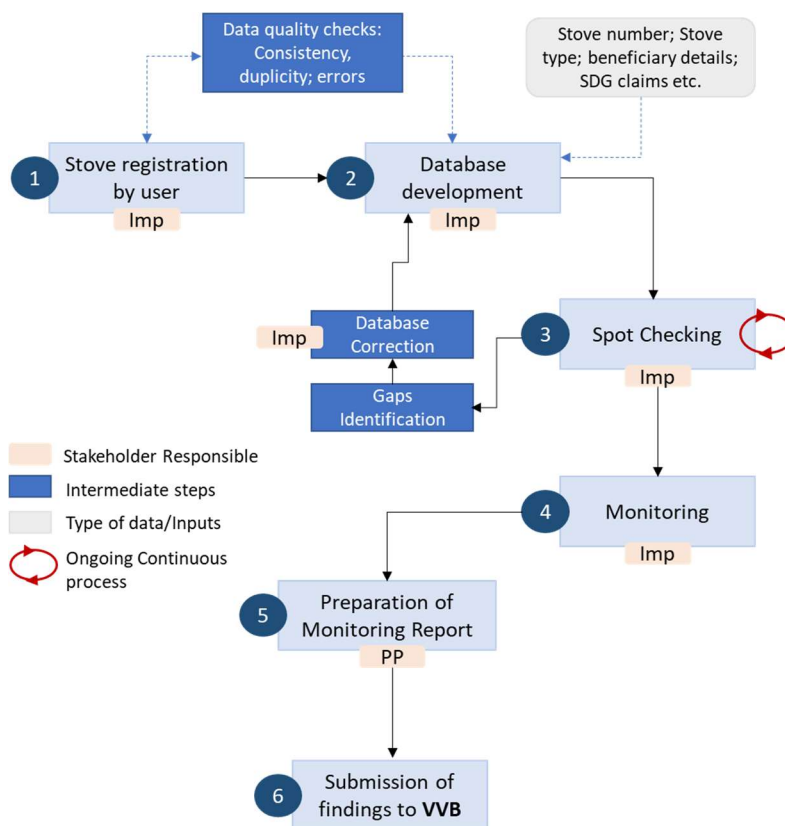
Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	Not required
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	10 years for all models of proposed stoves (Baldwin 8" Stove, Baldwin 10" Stove, Dishari Green Stove, Baldwin Double Pot, and Dinabandhu Stove).
Monitoring equipment	Not applicable
QA/QC procedures to be applied	PP conducts periodic audits of the stoves that will be distributed under the project activity to ensure their proper functioning throughout the project lifetime.
Purpose of data	Calculation of emission reductions
Calculation method	Not applicable
Comments	-

5.3 Monitoring Plan

The Monitoring Plan applied in this project involves several key elements that ensure that the PP and/or project Implementer have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Carbon Units (VCUs) following VCS methodology (VMR0006 version 1.1) on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- Data collection procedures
- Distribution and Monitoring Database
- Spot Checking of ICS (ongoing)
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

The below flow-chart illustrates the roles and responsibilities of the parties during the implementation of the monitoring plan for the project. In the below flowchart, the project implementer is abbreviated to “Imp” and can be the PP or another party authorized by the PP.



Below is the description of the above steps on the flow-chart.

- 1. Project Implementor: User stove registration:** Project implementer will collect/receive the necessary information requested in the registration process from the user. Server based

Mobile Application shall be used as means of collecting the information and for storing into the centralized database management system. Project Implementers' staff shall double check the accuracy of information provided, and request for field staff additional clarifications if needed. Following is the information that will be captured from each end-user at the time of registration of ICS during the project implementation, to be eligible for inclusion in the project activity:

- Unique identification alphanumeric ID for each ICS that will be distributed.
 - Date of distribution/installation of ICS.
 - Geographic area/address of recipient household.
 - GPS coordinates of the installed project device.
 - ICS Model/type of project technology distributed.
 - Quantity/number of project device that will be distributed.
 - End-User name and telephone number (if available).
2. **Project Implementor: Data logged into database:** Project implementer trained staff shall double check the information stored on the database and check for duplications and other entry errors. Any duplicate information shall be investigate and corrected or excluded from the database if it is a true duplicate entry.
3. **Project Proponent: Spot- checking (ongoing):** PP has a standard procedure for internal audit called spot-checking, where field staff randomly selects households and visit or contact the stove owners to cross-check the information stored on the database with the factual evidence in the field, to ensure proper functioning of the project stoves throughout the project lifetime and to ensure that the project stoves continue to work as efficient as specified by the manufacturer. Any inconsistencies found (e.g., change in the address of a user) gets updated on the database, and in case, ICS are found to be no longer in use, they get clearly marked as such and is excluded from emission reductions calculations.

During the spot check, if any of the project stoves was found to be in poor working condition like cracked/damaged or with missing metal parts, the stove owner repairs the stove as they have been trained during installation. If the stove owners are unsuccessful in repairing, the field staff informs the implementer to provide the necessary assistance and the missing metal parts are reported to the management of implementor who then either provides the parts from their existing stocks or request the same to project proponent who procured those from their suppliers. Stove owners are then provided with the missing parts by the implementing partner. Spot checking is a continuous ongoing process

4. **Project Implementor: Monitoring:** Project implementer will follow all the standard requirements to collect the necessary information needed for preparing the monitoring report.
5. **Project Proponent: Preparation of Report:** The PP will prepare the final monitoring report to be provided to the VVB for verification of emission reductions. A copy of the monitoring

report will remain with the PP.

The PP will coordinate and manage each project Implementer and assist them in implementing each element of the monitoring plan. The monitoring plan shall be elaborated in accordance with the Sampling Plan below. Also, regular audits conducted throughout the project lifetime ensures proper functioning of stoves and determination of the lifespan of the same.

Sampling Plan

As per the Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0, the sampling plan is the following:

a. Sampling design

Due to the large number of ICS envisioned to be distributed under the project, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling shall be undertaken for each batch of ICS. Samples will be drawn from ICS recorded in the project database administered by the project proponent.

I. Objectives and Reliability Requirements

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period.

Monitored Parameters:

No.	Monitoring parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,i,j}$ Number of project devices of type i and batch j operating during year y	Proportion of ICS still in operation	Proportion	At least once in two years.
2	$B_{y=1,new,i,j,survey}$ Quantity of woody biomass used by project devices in tonnes per device of type i . and batch j	Daily consumption of woody biomass per ICS	Mean value	Determined in the first year of project implementation

II. Sampling Method

The sampling method for the monitored parameters $N_{y,i,j}$ and $B_{y=1,new,i,j,survey}$ is Simple Random Sampling and samples will be randomly selected from the primary sampling units as illustrated

above. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

To determine the parameters, sampling will involve the following approaches (outcome in brackets):

$N_{y,i,j}$	Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)
$B_{y=1,new,i,j,survey}$	Interview with end user and estimate the daily consumption of woody biomass on project device (Measure the daily consumption of woody biomass on ICS)

Sample Size:

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using a 90 per cent confidence level, and a 10 per cent margin of error, random samples will be selected from each Primary Sampling Unit. The parameter number of stoves still in operation during the monitoring period as determined by the monitoring survey ($N_{y,j}$) will be estimated through sampling biennially and the parameter Quantity of woody biomass used by project devices in tonnes per device $B_{y=1,new,i,survey}$ shall be estimated through sampling within the first year of project installation. $B_{y=1,new,i,survey}$ can be coupled with $N_{y,j,j}$ for the first year. Of the two parameters to be monitored, $N_{y,i,j}$ is proportion/percentage parameter and $B_{y=1,new,i,survey}$ is a mean value parameter.

Proportion parameter ($N_{y,i,j}$)

To estimate the sample size for proportion parameters, the following equation³³ is used:

$$n \geq \frac{1.645^2 N p (1 - p)}{(N - 1) \times 0.1^2 \times p^2 + 1.645^2 p (1 - p)}$$

Where:

n	= Sample size
N	= Population Size (Total number of ICS)
p	= Expected proportion
1.645	= Representing 90% confidence interval (In the case of 95% confidence, 1.96 shall be used)

³³ Equation 1 of Appendix 3, Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities (Version 04.0)

0.1 = Represents 10% relative precision

Mean Parameter ($B_{y=1, \text{new}, i, \text{survey}}$)

For the purposes of determining sample size in the first monitoring period, the performance of ICS is characterized by the range of likely mean wood consumption and the likely values of SD relative to the mean, according to the type of ICS. The ICS models that are manufactured in modern factories tend to be very highly efficient (30-50% thermal efficiency) and have been designed to meet stringent efficiency specifications, so the standard deviation is expected to be relatively low.

To estimate the sample size for mean value parameter, the following equation³⁴ is used:

$$n \geq \frac{1.645^2 NV}{(N-1) \times 0.1^2 + 1.645^2 V}$$

Where:

$$V = \left(\frac{SD}{\text{mean}} \right)^2$$

n = Sample size

N = Population size (Total number of households/ICS)

mean = Expected mean

SD = Expected standard deviation

1.645 = Represents the 90% confidence required

0.1 = Represents the 10% relative precision

In case the sample size calculation returns a value of less than 30 samples, a minimum sample size of 30 would be chosen when the parameter of interest is a proportion. If the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the Student's t- distribution shall be used if the resulting sample size is less than 30.

The sample size to be determined via Student's t-distribution is as per the equation below³⁵:

³⁴ Equation 4 of Appendix 2, *Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities* (Version 04.0)

³⁵ Equation 38, *Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities*, pg. 46(version 04.0)

$$n = \left(\frac{t_{n-1} \times SD}{0.1 \times \text{mean}} \right)^2$$

Where t_{n-1} is the value of the t-distribution for 90% confidence when the sample size is n. Since the sample size is not known yet, the first step is to use the value for 90% confidence when the sample is large, i.e. 1.645 and then redefine the calculation.

The project proponent may choose to use the same sample to monitor more than one parameter. According to the Standard for sampling and surveys for CDM project activities and programme of activities, if there is more than one parameter to be estimated, then a sample size calculation should be done for each of them. Then either the largest number for the sample size is chosen as sampling effort with one common survey, or separate sampling efforts and surveys are undertaken for each parameter.

III. Sampling Frame

Since the end users shall always be households and geographical location is restricted to the project boundary it is therefore expected that the end user locations do not have influence on the parameters of interest that is $N_{y,i,j}$, & $B_{y=1, \text{new}, i, \text{survey}}$. Thus, these parameters can be assumed to be highly homogeneous regardless of how the end user group and distribution/installation location is defined. The Sampling Frame for individual parameters have been defined in the following paragraphs:

The sample frame refers to all the information sources on the Database. The cloud-based database management system via mobile application is the primary mechanism for data collection for newly distributed ICS and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the project. The sample size calculation to represent ICS population from the stored database for the Monitoring Survey follows the “Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities”; version 09.0.

The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error. Using, a 90 per cent confidence level, and a 10 per cent margin of error, random samples will be selected from the target population.

a. Sampling frame for proportion of ICS still in operation ($N_{y,j}$)

The first step is to identify the Primary Sampling Units.

Factors such as model of the ICS do not effect parameter $N_{y,j}$. Therefore Primary sampling unit in case of parameters $N_{y,j}$ is independent of model of stove.

b. Quantity of woody biomass used by project devices in tonnes per device ($B_{y=1,new,i,survey}$)

The Parameter $B_{y=1,new,i,survey}$ shall vary in accordance with its model. Hence for this parameter, the Primary Sampling Unit shall be defined as the group of ICSs of the same model. If two different ICS models being implemented in the same project – this will form two Primary Sampling Units. The value will be determined in the first year of the introduction of the devices through the monitoring survey.

b. Data to be collected:

(i) Field Measurements:

The table below summarizes field measurement data requirements.

Parameter	Timing (indicative)	Frequency as required by methodology	Methods to be applied	Comments on seasonal fluctuation
$N_{y,i,j}$	Monitoring will likely occur every 12 to 24 months	Biennially	Visits to the premises, visual inspection and interview with ICS end-user.	Not due to any seasonal fluctuation.
$B_{y=1,new,i,j,survey}$	Monitoring will likely occur within the first year after installation	First year of installation	Visits to the premises, visual inspection and interview with ICS end-user.	Seasonal fluctuation is not expected to affect this parameter. However, in case the survey results show a difference, the same shall be taken into account in ER estimation.

Soft copies of original stove purchase contracts, information collected from the registration or other means of acceptance by the users will be stored in the project office. A back-up of the project database will also be stored on an electric medium. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of credits for the project activity, whichever is later.

(ii) Quality Assurance/ Quality Control:

The project proponent will apply measures to ensure the required confidence/precision for each sampled parameter is met, allowing for non-response and the possible removal of outliers from the sample, as part of a Quality Control/Quality Assurance system.

The sampling plan has the following procedures in place to ensure good quality data. The project proponent will ensure that field personnel have reviewed, understood and have agreed to follow the monitoring plan procedures. A quality control and assurance strategy will be documented. The project proponent or a competent third party designated by the project proponent with the proper skills will train the monitoring personnel on how to properly survey households to prevent bias from interviewer.

The sample data for mean value parameter is continuous and therefore the presence of outliers is possible. To identify and address outliers for the parameter, outliers will be defined as those data points with values greater than three standard deviations from the mean of the sample. Data points identified as outliers will be examined further to correct for possible transcription and data entry errors but will be omitted from the analysis if no such administrative errors exist.

(iii) Analysis:

The project proponent will manage a project database that includes the end-user name, community/location, distribution/installation date and unique serial number data that can be directly attributable to each batch within the project, thereby allowing unambiguous determination of the emission reductions attributable to each project:

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values will then be factored into the emissions reduction calculations and result in the request for issuance of VERs. The stoves that are not in use will be excluded from emission reductions calculations and will not be counted towards the total number of ICS in operation during the monitoring period.

(iv) Implementation

Sampling for the purpose of emission reduction calculation and elaboration of the monitoring report will occur at the end of each monitoring period. This sampling will be conducted by trained personal from project proponent or an experienced third-party entity. The credentials and/or training materials for the sampling personal will be provided to the VVB at verification. The project proponent will be responsible for managing household data collection and entry into the project database. Appropriate record keeping procedures will be implemented to ensure that each monitoring period data set can be transparently attributed to its corresponding batch/project, preventing any occurrences of double counting. An internal review of the project database will be able to determine the current status of each batch — the duration of previous monitoring periods, the households delivering monitoring data, and current verification activities.

Assessment for Leakage

According to methodology VMR0006, version 1.1, section 8.3; Leakage shall be considered as default 0.95 in accordance with Section 5.4 of AMS-IL.G. therefore default value of leakage as 0.95 will be applied.

Monitoring Reporting

The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by those batches during the monitoring period.

APPENDIX A: LOCAL STAKEHOLDER CONSULTATION

Invitation via Email

Consultation meeting with local stakeholders for the project "Installation of High Efficiency Wood Burning Cookstoves in Bangladesh"

CQC_CSAT

To: Chandan Saha

Cc: Pooja Verma; Akhilesh Joshi; Tridip Goswami; Siddhartha Pakrashi; CQC_CSAT

Bcc: kamal.mig33@gmail.com; rahmanfahim9236@gmail.com; ppsl.shakib@gmail.com; sajidurrahmansiddiky@gmail.com; shakur.pranto2018@gmail.com; hasanbhuban85@gmail.com; ceo@bluebaybd.com; ashraf.islam@tetratetech.com; aamahmud@encotechbd.com; iftekhar.aziz@ap.averydennison.com; jtcdbd42@gmail.com; eemon441@gmail.com; ahmednasima576@gmail.com; atahsinr27ayasha@gmail.com; tbristy18@gmail.com; tbristy18@gmail.com; sanam000168@gmail.com; anushkaahmed1610@gmail.com; shaminsumas2@gmail.com; chowdhuryhafa90@gmail.com; rimondhali72@gmail.com; tasnimmuskan383@gmail.com; m.m.rashedhasan@gmail.com; sharfuddinahmed427@gmail.com; multitradebd@yahoo.com; shahidulalamdiu@gmail.com; eblmdhkmr@gmail.com; soniaazmain017@gmail.com; sharifulazad2@gmail.com; lito2002@gmail.com; ahmedrana000168@gmail.com; roxwptest@gmail.com; efath000168@gmail.com; roxtest@gmail.com; roxwptest@gmail.com; mahfuz643@gmail.com; baizid4432@gmail.com; sultan99878@gmail.com; farid44456@gmail.com; suza33456@gmail.com; gm.munnakhan@gmail.com; infohrdcbd@gmail.com; ushyang.pchowdhury@gmail.com; singhahwo@gmail.com; rasul_sdp@yahoo.com; rehabilitationcentrehrdc@gmail.com; hddbangladesh@gmail.com; pallavi@carboncheck.co.in; vikash@carboncheck.co.in; sanjaysingh@carboncheck.co.in; Madhurjya Gogoi; Sutrisna Khan; Kishore Das; Debarshi Das; Hemant Sharma; Gaurav Pandey

Sat 4/29/2023 11:00 PM

Non technical summary_Ban... 408 KB

FEEDBACK FORM-Installation... 18 KB

2 attachments (426 KB) Save all to OneDrive - C-Quest Capital Download all

Dear Sir / Madam,

C-Quest Capital BD Stoves 1 Pte Ltd with its implementation partner, as part of the requirement under the VCS project development process, are pleased to invite you to participate in the Local Stakeholders Consultation Meeting for the Project 'Installation of High Efficiency Wood Burning Cookstoves in Bangladesh' scheduled on 6th May 2023; at 11:00 AM to 02:00 PM in JRC Hall Room, BUET Graduates Club, House No. 67, Road No. 1, Banani, Dhaka-1213, Bangladesh.

These Projects aim to achieve sustainable development goals, by installing improved cookstoves (ICS) in households across Bangladesh. The project cookstoves have proven capacity to reduce fuel consumption and significantly reduce smoke by optimizing the combustion process while avoiding unnecessary heat loss. With the foresight of sustainable development, the proposed projects are expected to accelerate the shift from inefficient three-stone fire to more efficient cookstoves, resulting in not only emission reduction but also an improvement in the quality of life of millions of women and girls.

The proposed project will apply for carbon finance through a program of VERRA, the Verified Carbon Standard (VCS) to obtain Verified Carbon Units. It is in this context a Local Stakeholders Meeting has been organized to discuss the design of the project and its impact on sustainable development, as well as to receive valuable comments and suggestions from interested stakeholders.

Please find the program details below:

Agenda:

11:00 AM - 11:15 AM - Welcome address and introduction to Verified Carbon Standard

11:15 AM - 12:15 PM - Short Presentation on Project and its benefits to the stakeholders

12:15 PM - Onwards - Discussion & Feedback session

Those who would not be able to attend the in-person meeting may please join the Google Meet with the links given below.

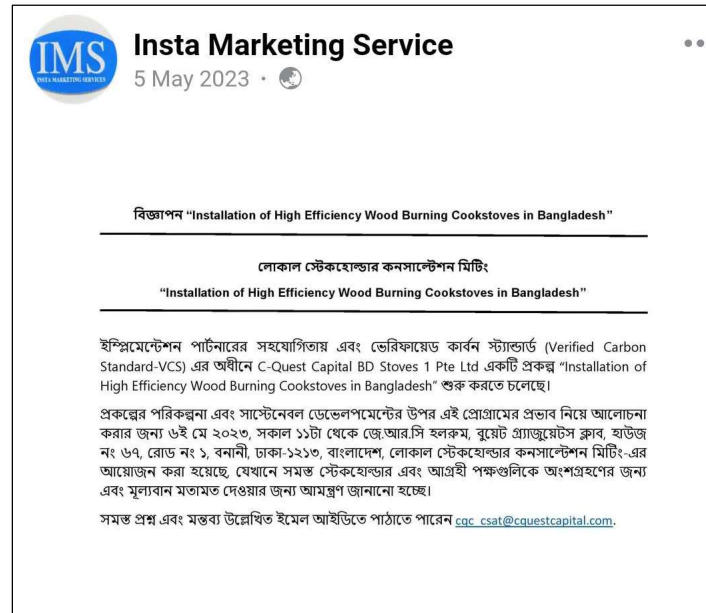
Google Meet link: <https://meet.google.com/twv-yseh-iz>

A non-technical summary and feedback form are attached to the email for your consideration. In case of any clarification or comment/suggestion regarding the project, you can contact us at cqc_csat@cquestcapital.com.

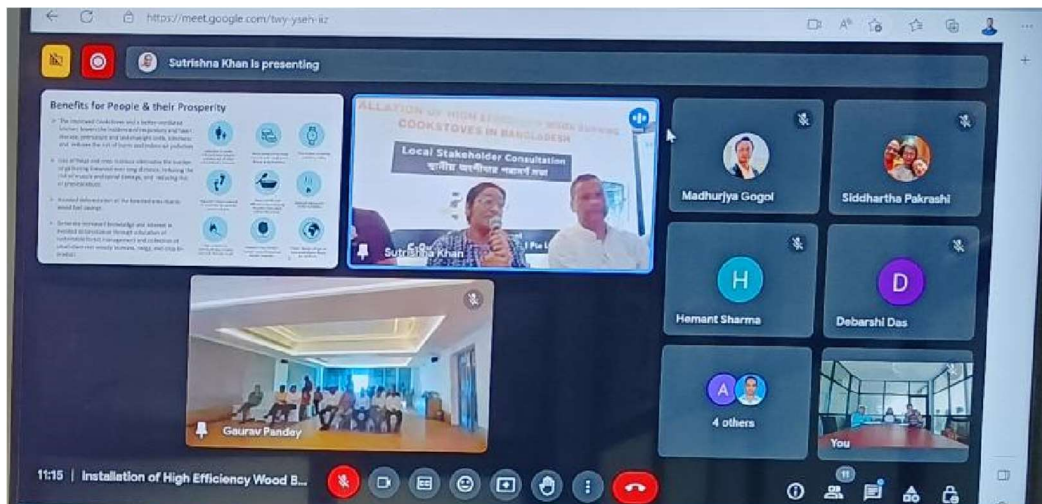
We look forward to your participation.

Thanks, and regards,
C-Quest Capital BD Stoves 1 Pte Ltd
CQC_CSAT@CquestCapital.com

Invitation via social media post³⁶



LSC Meeting (Online Snapshots)



³⁶ Insta Marketing Service | Facebook

LSC Meeting (Offline Snapshots)





Sample Evaluation Form

FEEDBACK FORM

Installation of High Efficiency Wood Burning Cookstoves in Bangladesh

Respondent Information:

Name: _____ Occupation: _____

Organisation Name: _____ Designation: _____

Contact number and Email: _____

1. What are the benefits you expect as a result of the Project Activity (PA) implementation?

- ☐ It will reduce the amount of wood required for cooking
- ☐ It will reduce smoke and indoor air pollution, and results in health benefits
- ☐ It will reduce the drudgery in the wood collection and cooking activities
- ☐ It will reduce costs and improve overall productivity and quality of life
- ☐ No benefits are envisaged

2. Do you think the project will be beneficial for the end users?

- ☐ Yes ☐ No

3. Are the project design and implementation plan clearly explained?

- ☐ Yes ☐ No

4. Are the project feedback and grievance redressal procedures clearly explained?

- ☐ Yes ☐ No

5. Do you think the project stoves will have a positive impact on the health of women and children?

- ☐ Yes ☐ No

6. Do you think the project will have a positive impact on the forests of Bangladesh?

- ☐ Yes ☐ No

7. Would you like to participate in this Programme?

- ☐ Yes ☐ No

8. Do you have any inputs, concerns, or suggestions on the project design or implementation plan?*

9. What is your impression of the project results, outcomes, and impacts (both positive/negative)? *

** You may use the back of this paper or request for additional sheets if the space is insufficient.*

ফিডব্যাক ফর্ম
Installation of High Efficiency Wood Burning Cookstoves in Bangladesh

উত্তরদাতার তথ্য:

নাম: _____ পেশা: _____

সংগঠনের নাম: _____ পদ: _____

যোগাযোগের নম্বর এবং ইমেল: _____

1. প্রকল্প কার্যকলাপ (পিএ) বাস্তবায়নের ফলে আপনি কী কী সুবিধা আশা করেন?
 - ☐ এটি রান্নার জন্য প্রয়োজনীয় কাঠের পরিমাণ হ্রাস করবে
 - ☐ এটি ধোঁয়া এবং ঘরের অভ্যন্তরের বায়ু দূষণ হ্রাস করবে এবং এর ফলে স্বাস্থ্যের উপকারিতা বৃদ্ধি পাবে
 - ☐ এটি কাঠের সংগ্রহ এবং রান্নার কাজের জটিলতা হ্রাস করবে
 - ☐ এটি খরচ হ্রাস করবে এবং সামগ্রিক উৎপাদনশীলতা এবং জীবনমানের মান উন্নত করবে
 - ☐ কোনও সুবিধা কল্পনা করা হয় না
2. আপনি কি মনে করেন যে প্রকল্পটি ব্যবহারকারীদের জন্য উপকারী হবে?
 - ☐ হ্যাঁ ☐ না
3. প্রকল্পের নকশা এবং বাস্তবায়ন পরিকল্পনার বিষয়ে কি এখানে স্পষ্টভাবে ব্যাখ্যা করা হয়েছে?
 - ☐ হ্যাঁ ☐ না
4. প্রকল্পের প্রতিক্রিয়া এবং অভিযোগ নিষ্পত্তির পদ্ধতি কি এখানে স্পষ্টভাবে ব্যাখ্যা করা হয়েছে?
 - ☐ হ্যাঁ ☐ না
5. আপনি কি মনে করেন যে প্রকল্পের চুলার ব্যবহার মহিলা ও শিশুদের স্বাস্থ্যের উপর ইতিবাচক প্রভাব ফেলবে?
 - ☐ হ্যাঁ ☐ না
6. আপনি কি মনে করেন যে এই প্রকল্পটি বাংলাদেশে অন্যান্যের উপর ইতিবাচক প্রভাব ফেলবে?
 - ☐ হ্যাঁ ☐ না
7. আপনি কি এই প্রোগ্রামে অংশগ্রহণ করতে চান?
 - ☐ হ্যাঁ ☐ না
8. প্রকল্পের নকশা বা বাস্তবায়ন পরিকল্পনা সম্পর্কে আপনার কি কোন মতামত বা পরামর্শ আছে? *
9. প্রকল্পের ফলাফল এবং প্রভাব (ইতিবাচক / নেতিবাচক উভয়) সম্পর্কে আপনার কি ধারণা? *

List of Offline Attendees

Name	Occupation/Organisation	Address (District/Organisation office address)
Md. Solrimon Al	Job	Bogura
Anubsh Roy	Job	Bogura
Md. Abdur Rahim	Job	Bogura
Ariful Hlam	Job	Bogura
Md. Zahangir Alam	Entrepreneur	Patuakheli
Md. Shenazul Haque	HDD	Kasba, Brahmanbarin
Mahbul Haque	HRDC	Dhaka
Aminur Rasul	CUP	Dhaka
Tahmin Aklir Dok	HRDC	Dhaka
Dr. Ushny Chaudhry	UPHCSDPH, DVCE	Dhaka
Dr. Suramoni Singho	HWO Energy Limited	Sylhet
M. Kansarazi	Business	-
Md. Belayet Hussain	Business	28, Layal street, Lalbage
Md. Ashmt UI Mhmd	Business	Dhaka
Abdul AI Mhmd	Business	Dhaka
Abdur Rahman Akoh	Business	Dhaka, Lalbage
Md. Alamin	Student	Dhaka, Lalbage
Md. Rabby	Student	Dhaka, Lalbage
Anika Tahsin	Student	Dhaka, Lalbagh
Begam Subhatara	Housewife	Lalbagh
Jannatul Dahsin	Student	Lalbagh
Rina Begum	Housewife	Lalbagh
Afrojanure Mim	Housewife	Lalbagh
Tasmin Rahman	Student	Lalbagh
Farzana Akhtar	Housewife	Lalbagh
M. Rahman Sagor	Dishari	Malbari hat, Sonatle, Bogira
Md. Emdadul Haque	HWO - Dhaka	Dhaka
Tahmid Iqbal	Tnnoctio	Dhaka

Md. Fahim Rahman	Hi Farm Limited	Dhaka
Pranto Mahmudul	D Link	Dhaka
Bijit Singha	Business	Dhaka
Md. Polash	Shah Solar	Cox's Bazar
Md. Shahin	PPSL	Dhaka
Md. Rajibul Hussan Shah	PPSL	Dhaka
Md. Hasan Zaman	Shah Solar	Cox's Bazar
Shaufel Azal	Fast Coi	159/A/1 A3 Prpur Dhaka
Jobyer Islam	Auditor	Paltam, Dhaka
Sarfuddin Ahmed	Dhaka	Lalbagh
Nasim Ahmed	Housewife	Lalbagh
Raisul Islam	Job	Lalbagh
Golos Mustaf (Munna)	Business	Lalbagh
Fahim Ahmed	Student	Lalbagh

List of Online Attendees

Name	Occupation/Organisation
Jawed Aslam	-
Nilanjana Banerjee	CQC-CSAT
Debarshi Das	CQC-CSAT
Jimreeves Donald	CQC-CSAT
Shahed Fazal	IP
Madhurjya Gogoi	CQC-Operation
Sutrishna Khan	CQC-CSAT
Vijay Kumar	CQC-CSAT
Tridip Kumar Goswami	CQC-CSAT
koganti Narendra Nath	CQC-CSAT
Siddhartha Pakrashi	CQC-Operation
Gaurav Pandey	CQC-Operation
Chandan Kr. Sah	CQC-CSAT

Manoj K Sarma	CQC-CSAT
Aryamann Sharma	CQC-CSAT
Hemant Sharma	CQC-CSAT
Aditya Dhar	Trainee Carbon Check

Stakeholder's Queries and Comments

Stakeholder's Comments and Questions	Project Proponent's Responses
You are saying households will make an economical saving by using your project cookstove, how is that possible?	First, we will be distributing the stove free of cost. Second, households do not need to buy firewood, because they can easily light the stove by using branches/twigs, leaf litter, crop biproduct etc. So, households can have a good economical saving from it.
How are you saying that your project is towards sustainable development?	Social, economic, and environment are the three pillars of sustainable development. The stoves we offer contribute to the environment by reducing wood consumption, are economically distributed free of cost, and the stoves are energy efficient, which means less cooking time, and allow women enough time for their own activities; thus, it fulfills sustainable development perspective.
What are the main changes you have made in your project stove to the traditional cookstove, that you are saying the efficiency is high?	The material we use and the design we create helps to trap heat inside the cookstove, which helps to reduce the heat leakage, that means fast cooking and less wood consumption, therefore efficiency is improved than the traditional cookstove.
How is the use of stove reducing the toxic gas emission?	For traditional cookstoves, wood logs are used, and incomplete combustion may occur, causing emission of Carbon Monoxide. Which is harmful for the environment, and human body. But in our project stove only branches/twigs, leaf litter, crop biproduct etc. will be used as fuel, that will combust fully inside the stove, that will not produce toxic gases like Carbon Monoxide. In other words, as

	the wood consumption will be less in the stove so emission will be reduced.
How do you explain environmental pollution to the end-user?	With the help of IP, we will organize training or meeting programs in various locations in rural places, to make them about the pollution, and how that is harming our body and environment.
How is project stove helping to improve the health condition of the end-user?	The end-user especially women who use traditional cookstove suffer from eye irritation, breathing problems, lung problem, cancer etc. due to long exposure of smoke and toxic gases. In our project stove toxic gas and smoke emission will be reduced. So that will help them to improve their health.
The delta areas of Bangladesh are flooded during monsoon. At that time the people there changed their place and went to higher land. What will you do to understand what kind of stove will be distributed to such families?	We will do a survey there in delta areas, to understand people's needs. Then we will decide what will be the right type of cookstove for that place.
If any end-user face problem with the stove, how you will help them to sort the problem?	Once our project is implemented our ground team will be surveying on a regular basis, to know the feedback of the end-user, if any problem raised by end-user our ground team will act on that. Otherwise, we have grievance policy, where end-user can directly connect us to get solution on their problems.

APPENDIX B: SCOPE 3 EMISSION STATEMENT

Public Notice for VCS Project 4312

Installation of High Efficiency Wood Burning Cookstoves in Bangladesh

This is to bring to your kind notice that C Quest Capital BD Stoves 1 Pte Ltd is implementing a grouped project titled, "Installation of High Efficiency Wood Burning Cookstoves in Bangladesh" (VCS ID 4312). It involves the distribution of improved cook stoves (ICS) to households in Bangladesh. For this purpose, we are procuring Baldwin Stoves from Entrust Products, Dinabandhu Stove Metal Parts from Modern MiniSuperMarket (MMSM), and Dishari Green Stoves from Dishari Enterprise Ltd. (DEL). We have also involved various local implementation partners for the distribution of project stoves in Bangladesh.

Verified Carbon Units (VCUs) may be issued for the greenhouse gas emission reductions and removals for which C Quest Capital BD Stoves 1 Pte Ltd will be claiming carbon credits under VERRA. The ownership of these credits lies exclusively with C Quest Capital BD Stoves 1 Pte Ltd.

This notice intends to apprise you about the project, to avoid any potential risk of double claiming Scope 3 emissions under this grouped project.

Public notice³⁷ published to avoid double claiming of Scope 3 emissions

³⁷ <https://cquestcapital.com/latest/public-notice/>